

Repairo Architecture & Workflow Docs

System Architecture, Process Workflow & Security Architecture Specification

SARVAM STARTUP PROGRAM BACKED

1. System Overview & Core Mission

Repairo is an automated API dependency maintenance platform built for modern engineering teams. When third-party API vendors (OpenAI, Google Gemini, Stripe, Supabase) publish breaking changes, deprecate parameters, or release new major SDK versions, Repairo automatically detects the breaking contract changes, scans customer codebases, and generates deterministic, zero-retention GitHub Pull Requests to fix the broken code.

2. End-to-End Technical Process Workflow

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[Step 1: Vendor Spec Poller] → [Step 2: Repository Scanner]
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[Step 4: Automated PR Engine] ← [Step 3: Volatile RAM AST Engine (~24ms purge)]
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- **Step 1: OpenAPI Spec Polling & Contract Diffing:** Background poller microservices running on Railway continuously fetch updated OpenAPI YAML/JSON schema definitions from vendor changefeeds. The Contract Spec Differ compares the old vs new spec and generates a Breaking Change Matrix.
- **Step 2: Repository Scanning:** When a breaking change is registered, Repairo triggers a scan across connected user repositories to search for call-site patterns matching deprecated vendor methods (e.g. `openai.ChatCompletion.create`).
- **Step 3: Volatile RAM AST Transformation Engine:** Repairo streams source code into an isolated, volatile RAM buffer (Zero-Retention Vault). The AST engine (TypeScript Compiler API, Babel, Tree-sitter) parses AST nodes and rewrites call-sites. The memory is flushed in ~24ms. Zero code touches disk.
- **Step 4: Automated Verification & GitHub PR Generation:** Verified code patches are pushed to a clean git branch, and an automated Pull Request is opened on GitHub containing unified git diffs and vendor changelog references.

3. Security & DevSecOps Architecture (Domain Lead: Rehan Shaikh)

- **Zero-Retention Memory Vault:** All code parsing occurs strictly in volatile RAM with zero disk persistence and ~24ms memory purging.
- **SBOM CVE Dependency Scanning:** Automated GitHub Actions workflows (`.github/workflows/sbom-scan.yml`) run Trivy/Grype scanners on patched dependencies.
- **Infrastructure Hardening:** Next.js 16 hosted on Vercel Edge, background workers on Railway VPC, and AES-256 encrypted OAuth secrets.